A Learnable Feature Preprocessing Front-End Based Network for SAR Ship Classification

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***Abstract*—This paper proposes LFE-Net, a network with a learnable preprocessing front-end for Synthetic Aperture Radar (SAR) image ship classification. The method makes the extraction of scattering and texture features learnable through parameterization, enabling end-to-end optimization and moving beyond fixed, handcrafted preprocessing steps. Experiments on the OpenSARShip 2.0 dataset show that LFE-Net achieves significant accuracy improvement over various mainstream networks, validating the effectiveness of the learnable preprocessing mechanism. This work offers a new approach for adaptive feature extraction in SAR images.**

***Index Terms*—learnable preprocessing, ship classification, Synthetic Aperture Radar(SAR), scattering features,texture features.**

I. Introduction

Synthetic Aperture Radar (SAR) is a mature and powerful remote sensing technology capable of imaging the Earth's surface day and night under all weather conditions, playing an indispensable role in maritime surveillance [1]. Currently, SAR is widely applied in maritime ship monitoring and environmental management [2]. As a critical technology for maritime surveillance, SAR ship classification has consistently been an active area of research within the remote sensing community. Recent research on SAR image ship classification mainly focuses on preprocessing the polarimetric features of SAR images to generate new images, which are then used as input to neural networks to provide the network with rich prior knowledge. Xie et al. [3] constructed two types of pseudo-RGB images comprising polarimetric and texture features from dual-polarized SAR images by utilizing theories such as polarimetric entropy, degree of polarization, and local binary patterns (LBP) [4], and used them as network input, significantly improving classification accuracy. Similarly, Chen et al. [5] constructed two pseudo-RGB images composed of different polarimetric features from dual-polarized SAR images using theories like Pauli decomposition and used them as model input, achieving better performance compared to using the raw, unprocessed images directly as network input. However, such fixed preprocessing is neither efficient nor integrated into network learning, limiting feature adaptability. To address this, we propose LFE-Net, which incorporates learnable scattering and texture preprocessing front-ends into end-to-end training for enhanced performance.

The main contributions of this paper are as follows:

1) A learnable preprocessing front-end that parameterizes the traditionally fixed preprocessing steps, enabling end-to-end optimization for SAR ship classification.
2) A novel texture feature extraction methodthat computes texture using actual pixel values, jointly modeling both the relationship between the center pixel and its neighbors and the relationships among neighboring pixels themselves.

The structure of this paper is as follows: Section II briefly introduces the proposed method; Section III presents and analyzes the experimental results; Section IV provides the conclusion and discussion.

II. PROPOSED METHOD

The proposed method in this paper utilizes dual-polarized SAR image amplitude data. This section primarily introduces the learnable scattering feature preprocessing front-end, the learnable texture feature preprocessing front-end, and the feature extraction method.

A. Learnable Scattering Feature Preprocessing Front-End

For dual-polarized SAR image amplitude data, this paper proposes a learnable design based on Pauli decomposition. Pauli decomposition is a classical polarimetric target decomposition method. The formulas for its three components are:

$$P_1 = (I_{VV} + I_{VH})/\sqrt{2}, \tag{1}$$

$$P_2 = (I_{VV} - I_{VH})/\sqrt{2}, \tag{2}$$

$$P_3 = I_{VH}/\sqrt{2}. \tag{3}$$

Equations (1), (2), and (3) use I_{VV} and I_{VH} to denote the VV and VH channels in the dual-polarized data, respectively.

These three components reflect the overall scattering, double-bounce scattering and surface scattering characteristics of ship targets,respectively. They can effectively capture the structural variations of ships, aiding the model in more accurately distinguishing between different target types based on their physical scattering characteristics [5].

Pauli decomposition employs fixed linear combination coefficients.To achieve learnability, we transform these three fixed basis vectors into learnable parameters. Specifically, we define three learnable two-dimensional basis vectors $\mathbf{v}_i \in$

$\mathbb{R}^2 (i = 1,2,3)$ and initialize them with physically meaningful values,$\mathbf{v}_1^{(0)} = [1/2, 1/2]$, $\mathbf{v}_2^{(0)} = [1/2, -1/2]$,$\mathbf{v}_3^{(0)} = [0, 1/2]$.Here,$\mathbf{v}_1^{(0)}$,$\mathbf{v}_2^{(0)}$,$\mathbf{v}_3^{(0)}$ correspond to the fixed coefficient combinations for P_1,P_2,P_3in the Pauli decomposition, respectively, and serve as the initial values for network training. During optimization,$\mathbf{v}_i$ are updated within a constrained range around their initial values, preserving physical interpretability while enabling adaptive, end-to-end learnable feature extraction.

To enable adaptive learning while preserving physical interpretability, we impose a soft constraint on the parameters. After each parameter update, each component of the basis vector parameter $\mathbf{v}_i$ is constrained to remain within a $\pm\varepsilon$ neighborhood of its initial value $\mathbf{v}_i^{(0)}$. This constraint ensures that the learning process does not deviate excessively from the physically meaningful Pauli space, thereby balancing the model's flexibility with the feature's physical interpretability.

Given a dual-polarized input image of size $H \times W$ the VV and VH intensity values at each pixel position (m,n) form a two-dimensional vector $\mathbf{u}(m,n) = [I_{VV}(m,n), I_{VH}(m,n)]^\top \in \mathbb{R}^2$. This module generates three scattering feature response values by computing the dot product between this vector and each learnable basis vector, as expressed in (4):

$$\mathbf{F}_i(m,n) = \mathbf{v}_i \cdot \mathbf{u}(m,n), i = 1,2,3 \tag{4}$$

By traversing all pixel positions (m,n) in the image, we obtain three two-dimensional response maps $\mathbf{F}_1, \mathbf{F}_2, \mathbf{F}_3 \in \mathbb{R}^{3\times H\times W}$. Stacking them along the channel dimension yields the final three-channel scattering feature map $\mathbf{F}_{scatter} = [\mathbf{F}_1, \mathbf{F}_2, \mathbf{F}_3]^\top \in \mathbb{R}^{3\times H\times W}$. The entire module is computationally efficient, involving only simple dot product operations, and all parameters are optimized via gradient backpropagation.

B. Learnable Texture Feature Preprocessing Front-End

Common methods for extracting texture features include LBP and its variants [6], [7],we employs actual pixel value differences to extract texture features. For dual-polarized SAR images, they are processed into two-channel texture feature maps by computing the difference relationships between the central pixel and its neighbors, based on a modified TA module [8] originally designed for fully polarimetric data. Furthermore, the local texture differences within the neighborhoods of different polarized SAR images are modeled and weighted to serve as the third channel of input to the visual encoder.

Specifically,The front-end takes a dual-polarized SAR image of size $H \times W$ as input and outputs three texture feature maps of the same size: $\mathbf{T}_{VV}$,$\mathbf{T}_{VH}$,and $\mathbf{T}_S$.

To generate the texture feature maps $\mathbf{T}_{VV}$ and $\mathbf{T}_{VH}$,the module first extracts a 3×3 neighborhood window at each pixel position. For a position (i,j),let the vector formed by all pixel values within its neighborhood window be denoted as $\mathbf{X}_c^{(i,j)} \in \mathbb{R}^9$, where the subscript $c \in \{VV, VH\}$ indicates the polarization channel. The fifth element of each vector is the central pixel value of the corresponding channel, i.e.,$X_{VV,5} = \mathbf{I}_{VV}(i,j)$ and $X_{VV,5} = \mathbf{I}_{VV}(i,j)$. Let $\mathbf{b}_c^{(i,j)} = (x_{c,5}, x_{c,5}, ..., x_{c,5})^\top \in \mathbb{R}^9$ represent the vector constructed by repeating the central pixel value.

First, the absolute difference vector between the neighboring pixels and the central pixel in each channel is calculated:

$$\mathbf{d}_c^{(i,j)} = \left| \mathbf{x}_c^{(i,j)} - \mathbf{b}_c^{(i,j)} \right| \tag{5}$$

Then, the difference vector is adaptively modulated by a learnable channel-specific scaling factor $\alpha_c \in \mathbb{R}$:

$$\widetilde{\mathbf{d}}_c^{(i,j)} = \alpha_c \cdot \mathbf{d}_c^{(i,j)} \tag{6}$$

To fuse dual-polarization information and enhance the texture differences, the two modulated difference vectors are summed, and a weight mask is generated using a learnable global parameter $\beta \in \mathbb{R}$ and the hyperbolic tangent function:

$$\mathbf{m}^{(i,j)} = 1 - \tanh\left(\beta \cdot \left(\widetilde{\mathbf{d}}_{VV}^{(i,j)} + \widetilde{\mathbf{d}}_{VH}^{(i,j)}\right)\right) \in \mathbb{R}^9 \tag{7}$$

Then, the original neighborhood pixel vector is re-weighted using the mask from (7):

$$\widetilde{\mathbf{x}}_c^{(i,j)} = \mathbf{m}^{(i,j)} \odot \mathbf{x}_c^{(i,j)} \tag{8}$$

In (8) , $\odot$ denotes element-wise multiplication.Finally, the re-weighted features are linearly combined via two independent sets of learnable 9-dimensional weight vectors $w_c \in \mathbb{R}$ to obtain the texture feature value at position (i,j):

$$\mathbf{T}_c(i,j) = \left(\widetilde{\mathbf{x}}_c^{(i,j)}\right)^\top \mathbf{w}_c \tag{9}$$

By iterating over all positions in the image, the texture feature maps $\mathbf{T}_{VV}$ and $\mathbf{T}_{VH}$ are generated. Additionally, to capture structural relationships, a third feature map $\mathbf{T}_S$ is generated by modeling all pixel-pair similarities within the same neighborhood.

For the same 3×3 neighborhood window at position (i,j),similarity matrices are computed separately for the VV and VH channels. For a channel c, its similarity matrix $\mathbf{S}_c^{(i,j)} \in \mathbb{R}^{9\times 9}$ has elements $\mathbf{S}_c^{(i,j)}(k.l)$ representing the similarity between the k-th and l-th pixels in the neighborhood, defined as:

$$\mathbf{S}_c^{(i,j)}(k,l) = \exp\left(-\gamma_c \cdot (x_{c,k} - x_{c,l})^2 - \eta \cdot r_{k,l}\right) \tag{10}$$

In (10),$x_{c,k}$ and $x_{c,k}$ are the intensity values of the k-th and the l-th pixel in channel c,respectively, and $r_{k,l}$ is the fixed Euclidean distance between pixel k and l in the 3×3 grid. The parameter $\gamma_c \in \mathbb{R}$ controls the sensitivity of channel c to intensity differences,and $\eta \in \mathbb{R}$ is a shared spatial distance penalty coefficient.

The similarity matrices from the two channels are then fused to obtain a comprehensive similarity matrix $\mathbf{S}_f^{(i,j)} \in \mathbb{R}^{9\times 9}$:

$$\begin{aligned} \mathbf{S}_f^{(i,j)} &= \lambda \cdot \mathbf{S}_{VV}^{(i,j)} + (1-\lambda) \cdot \mathbf{S}_{VH}^{(i,j)} \\ &+ \mu \cdot \left(\mathbf{S}_{VV}^{(i,j)} \odot \mathbf{S}_{VH}^{(i,j)}\right) \end{aligned} \tag{11}$$

In (11),$\lambda \in \mathbb{R}$ and $\mu \in \mathbb{R}$ are learnable fusion parameters used to balance the contributions of different polarimetric information and to reinforce dual-polarization consistency.

After obtaining $\mathbf{S}_f^{(i,j)}$, the pixel pairs are categorized into three levels based on their distance $r_{k,l}$: nearest neighbors $(r_{k,l} = 1)$,diagonal adjacency $(r_{k,l} = \sqrt{2})$,and long-range $(r_{k,l} > \sqrt{2})$. The average similarity within each level m is calculated as:

$$\bar{\mathbf{s}}_n^{(i,j)} = \frac{1}{|\mathbf{G}_n|} \sum_{(p,q)\in\mathbf{G}_n} \mathbf{S}_f^{(i,j)}(p,q) \tag{12}$$

In (12), $\mathbf{G}_n$,denotes the set of pixel pairs belonging to the m-th level, and $|\mathbf{G}_n|$ is the size of this set. Finally, the local feature value within the neighborhood of position (i,j) is determined by the weighted sum of these three level features:

$$\mathbf{T}_S(i,j) = \sum_{n=1}^{3} \widetilde{\phi}_n \cdot \bar{\mathbf{S}}_n^{(i,j)} \tag{13}$$

In (13), the weight $\widetilde{\phi}_n$ is obtained by normalizing a learnable parameter vector $\boldsymbol{\phi} = (\phi_1, \phi_2, \phi_3)^\top \in \mathbb{R}$ via the Softmax function, i.e., $\widetilde{\phi}_n = exp(\phi_n) / \sum_{j=1}^{3} exp(\phi_j)$,ensuring that the sum of the weights equals 1.

In summary, this texture feature preprocessing front-end transforms the input dual-polarized SAR image into a three-channel texture feature map $\mathbf{F}_{texture} = [\mathbf{T}_{VV}, \mathbf{T}_{VH}, \mathbf{T}_S]^\top \in \mathbb{R}^{3\times H\times W}$ through its learnable parameters. All parameters are optimized via gradient back-propagation during end-to-end training, enabling the texture extraction process to adapt to specific tasks and data distributions.

C. Feature Extraction and Classification

The overall architecture of the model is illustrated in Fig.1. To avoid introducing a massive number of parameters and increasing model complexity, we employ a layer-wise transfer learning approach, selecting the Swin Transformer [9] pre-trained on the ImageNet dataset as the backbone network. The Swin Transformer is an improvement over the original

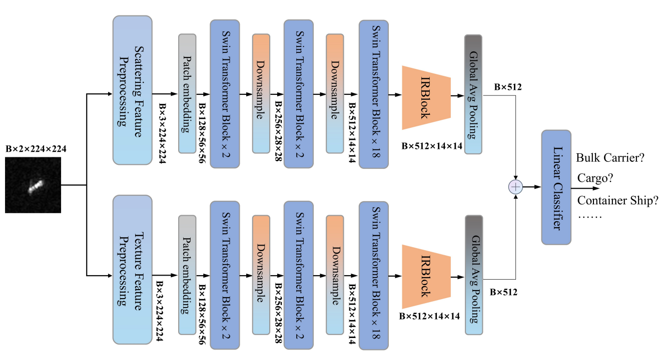

Fig. 1. Framework of LFE-Net

Transformer [10] .

To further reduce the number of parameters, we select the first three layers of the Swin Transformer-Base network as the visual encoder. Simultaneously, to further deepen the features, we choose to add an inverted residual block (IRBlock) after the visual encoder and make certain modifications to this IRBlock. The activation function used in the original IRBlock is ReLU6, which clips negative values and positive values exceeding 6, potentially leading to the loss of important polarimetric feature information [3]. Since the scattering feature extraction process can generate certain negative values, and to avoid the "dying neuron" problem, we replace ReLU6 with Leaky ReLU, which allows a small, non-zero gradient for negative values.

Furthermore, in terms of network architecture, the scattering and texture features undergo separate feature extraction, followed by an element-wise summation fusion via a global pooling operation. This approach incurs minimal computational overhead and avoids the performance degradation associated with simple decision-level fusion. Finally, the fused features are fed into a classifier for the final classification.

III. EXPERIMENTS AND DISCUSSION

A. Dataset and Experimental Setup

The OpenSARShip2.0 dataset [11] provides comprehensive ship chips. Since our modules are designed for amplitude data, we choose to use the amplitude-calibrated data from OpenSARShip2.0, specifically from its "Patch-Cal" folder.

Following the common dataset settings for ship classification in [3], [12], the six-class experiment uses the following categories: bulk carriers, cargo, container ships, fishing, general cargo, and tankers. The data was randomly selected and then partitioned in a 7:3 ratio between the training set and the validation set.

Our Experiments are implemented in PyTorch on an NVIDIA RTX 4090 GPU with CUDA 11.8. The learning rate is set to 1e-5, and the weight decay is 3e-4.The random seed is set to 42. The model is trained for 40 epochs with a batch size of 16. Logarithmic and Z-score normalization are applied to the features before they are fed into the network to mitigate the impact of numerical disparities in the raw data. We set ε = 0.15 (i.e.,15% of the initial value) and enforce the constraint by clipping each component after every gradient update.

B. Evaluation Criteria

This paper employs several commonly used classification evaluation metrics, including accuracy, precision, recall, and the F1-score. Their mathematical expressions are: Accuracy $= (TP+TN)/(TP+TN+FP+FN)$, Precision $= TP/(TP+FP)$, Recall $= TP/(TP+FN)$, F1 $= 2 \times (\text{Recall} \times \text{Precision})/(\text{Recall} + \text{Precision})$.

Here, TP denotes the number of correctly predicted positive samples, TN denotes the number of correctly predicted negative samples, FN denotes the number of incorrectly predicted negative samples, and FP denotes the number of incorrectly predicted positive samples.

C. Experimental Results and Analysis

The confusion matrix on the six-class validation set after model training is shown in TABLE I. TABLE I shows that our model achieves good performance on most categories, particularly on Fishing, though General cargo remains challenging due to its relatively small sample size, lower

TABLE I
SIX-CLASS CLASSIFICATION CONFUSION MATRIX OF LFE-NET.

True \ Predicated	Bulk Carrier	Cargo	Container Ship	Fishing	General cargo	Tanker	**Recall(%)**
Bulk Carrier	140	22	19	0	6	9	71.43
Cargo	30	306	9	1	22	67	70.34
Container Ship	21	22	184	0	18	8	72.73
Fishing	0	8	0	22	1	4	62.86
General cargo	4	27	7	0	17	15	24.29
Tanker	15	84	24	0	19	220	60.77
Precision(%)	66.67	65.25	75.72	95.65	20.48	68.11	**Accuracy=65.80%**
F1 Score(%)	68.97	67.70	74.19	75.86	22.22	64.23	

TABLE II
QUANTITATIVE RESULTS OF DIFFERENT METHODS ON OPENSARSHIP2.0.

Method	Recall(%)	Precision(%)	F1(%)	Accuracy(%)
VGG-11 [13]	57.88	56.94	56.85	57.88
ResNet-18 [14]	55.37	54.28	54.07	55.37
Swin Transformer [9]	49.67	52.40	45.90	49.67
Swin Transformer V2[15]	53.07	52.26	50.88	53.07
DenseNet-121 [16]	54.63	54.63	53.53	54.63
EfficientNetV2 [17]	48.85	44.75	49.84	48.85
Wang et al. [18]	56.77	54.33	54.05	56.77
HDSS-Net [19]	58.13	57.86	57.99	58.13
LFE-Net(S*)	61.73	62.25	61.57	61.73
LFE-Net(T*)	63.58	63.99	63.57	63.58
LFE-Net	65.80	66.65	66.03	65.80

image resolution, and the lack of distinctive features, unlike Fishing which has prominent characteristics for easy discrimination. Moreover, we did not assign a higher loss weight to this category during training. To benchmark our model, we compare it with several mainstream networks under the same experimental setup: VGG-11 [13], ResNet-18 [14], Swin Transformer [9],Swin TransformerV2 [15],DenseNet-121 [16],EfficientNetV2 [17],Wang et al [18],HDSS-Net [19]. The quantitative results of all methods are presented in TABLE II.

In the comparative experiments, we trained different algorithms under the same conditions and conducted an ablation study on LFE-Net. Here, S* denotes the result of training using only the polarimetric feature branch, and T* denotes the result of training using only the texture feature branch. It can be observed that our method achieves superior performance, whether using the complete network or individual branches for training. Taking VGG-11 as an example, our method achieves a performance improvement of 7.92% in accuracy, thereby validating the effectiveness of our approach.

To validate the effectiveness of the learnable parameters, we conducted an ablation experiment where these parameters in the preprocessing front-end were fixed. As shown in TABLE III, fixing them leads to a decrease in all classification metrics. This demonstrates that incorporating the learnable front-end

TABLE III
COMPARATIVE EXPERIMENT OF LEARNABLE PREPROCESSING AND FIXED PARAMETER PREPROCESSING.

Fixed-parameter module		Recall(%)	Precision(%)	F1(%)	Accuracy(%)
$F_{scatter}$	$F_{texture}$				
	√	64.69	65.35	65.02	64.69
√		64.91	65.63	65.12	64.91
√	√	63.51	64.49	63.84	63.51
		65.80	66.65	66.03	65.80

into training enables the model to learn a superior feature extraction strategy, resulting in a performance gain of over 2% compared to using fixed parameters. Finally, we present the changes of the learnable parameters set in the paper before and after training, and the results are shown in Table IV.

TABLE IV
INITIAL AND LEARNED PARAMETERS (SPLIT INTO TWO COLUMNS TO REDUCE HEIGHT).

Left group			Right group		
Parameter	**Initial**	**Learned**	**Parameter**	**Initial**	**Learned**
v_{11}	0.707107	0.672123	$\bar{w}_{\mathrm{VV}}$	1.000000	0.956361
v_{12}	0.707107	0.637560	$\bar{w}_{\mathrm{VH}}$	1.000000	0.936117
v_{21}	0.707107	0.772971	β	1.000000	0.968215
v_{22}	-0.707107	-0.597613	α_{VV}	1.000000	0.935686
v_{31}	0.000000	-0.016716	α_{VH}	1.000000	0.917941
v_{32}	0.707107	0.657773	γ_{VV}	1.000000	0.577891
ϕ_1	0.333333	0.393900	γ_{VH}	1.000000	0.451588
ϕ_2	0.333333	0.298400	η	0.500000	0.436118
ϕ_3	0.333333	0.307700	λ	0.500000	0.246797
			μ	0.500000	0.425083

IV. CONCLUSION

This paper presents LFE-Net, which introduces learnable preprocessing front-ends for scattering and texture features in SAR ship classification. By replacing fixed formulas with parameterized, end-to-end trainable modules, LFE-Net adaptively extracts more discriminative features. Experiments on OpenSARship2.0 validate its effectiveness, showing clear improvements over mainstream methods.

Although LFE-Net has achieved certain performance improvements in SAR image classification, further in-depth research into feature preprocessing methods and validation of its generalizability on broader datasets are still required.

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